



Git

Hands-On 4





Description:

This hands-on lab covers merge conflict resolution when concurrent updates to the master branch and a feature branch result in conflicting modifications. You will learn how to create a conflict, visualize it, use a 3-way merge tool to resolve it, and clean up the repository afterwards.

WorkFlow

Step 1: Create Diverging Changes in Branch and Master

Verify Clean State: Check that your master branch is in a clean working state:

git status

Create Branch and Add File: Create a new branch named GitWork and switch to it, then create a file named hello.xml:

git checkout -b GitWork
touch hello.xml

Update Branch File and Commit: Update the content of hello.xml within the branch, check the status, and commit the changes:

echo "<note>Branch content update</note>" >> hello.xml
git status
git add hello.xml
git commit -m "Add hello.xml in GitWork branch"

Switch to Master and Create Conflict: Switch back to master, create a file named hello.xml with different content, and commit it to master:

git checkout master
touch hello.xml
echo "<note>Master conflicting content update</note>" >> hello.xml
git add hello.xml
git commit -m "Add conflicting hello.xml in master branch"



WorkFlow

Step 2: Detect and Visualize Conflicts

View Commit History: Observe the logs across all branches to see the divergent history:

git log --oneline --graph --decorate --all

Check Differences: Use git diff or launch the P4Merge tool to inspect the differences between master and the branch:

git diff master GitWork

git mergetool

Step 3: Resolve Conflicts and Complete Merge

Attempt Merge: Merge the GitWork branch into master to trigger the conflict markers:

git merge GitWork

Resolve via 3-Way Merge Tool: Use your configured 3-way merge tool to interactively resolve the conflicts:

git mergetool

Commit Resolution: Once the conflict is resolved, commit the changes to the master branch:

git commit -m "Resolve merge conflict between master and GitWork"



WorkFlow

Step 4: Cleanup and Verification

Update .gitignore and Status: Check the repository status, add any generated backup files (e.g., .orig) to your .gitignore file, and commit the update:

```
git status  
echo "*.orig" >> .gitignore  
git add .gitignore  
git commit -m "Add backup files to .gitignore"
```

List Branches and Delete Merged Branch: List available branches and delete the merged GitWork branch:

```
git branch -a  
git branch -d GitWork
```

Final Log Inspection: Review the final repository log structure:

```
git log --oneline --graph --decorate
```



Git Bash & Output

```
MINGW64:/c/Users/AADI/GitDemo

AADI@Aadi MINGW64 ~
$ cd GitDemo

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git checkout -b GitWork
touch hello.xml
Switched to a new branch 'GitWork'

AADI@Aadi MINGW64 ~/GitDemo (GitWork)
$ echo "<note>Branch content</note>" > hello.xml
git status
git add hello.xml
git commit -m "Add hello.xml in GitWork branch"
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[GitWork b96402d] Add hello.xml in GitWork branch
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml

AADI@Aadi MINGW64 ~/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

AADI@Aadi MINGW64 ~/GitDemo (master)
$ echo "<note>Master content</note>" > hello.xml
git add hello.xml
git commit -m "Add hello.xml in master branch with different content"
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[master f987618] Add hello.xml in master branch with different content
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
* f987618 (HEAD -> master) Add hello.xml in master branch with different content
| * b96402d (GitWork) Add hello.xml in GitWork branch
|/
* 863caa6 Add feature.txt in GitNewBranch
* 7392b8c (origin/master) Add gitignore file to ignore log files and folders
* 088c41c Welcome

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
```

```
MINGW64:/c/Users/AADI/GitDemo

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
* f987618 (HEAD -> master) Add hello.xml in master branch with different content
| * b96402d (GitWork) Add hello.xml in GitWork branch
|/
* 863caa6 Add feature.txt in GitNewBranch
* 7392b8c (origin/master) Add gitignore file to ignore log files and folders
* 088c41c Welcome

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

AADI@Aadi MINGW64 ~/GitDemo (master|MERGING)
$ git add hello.xml
git commit -m "Resolve merge conflict between master and GitWork"
[master 70d781e] Resolve merge conflict between master and GitWork

AADI@Aadi MINGW64 ~/GitDemo (master)
$ echo "*.orig" >> .gitignore
git add .gitignore
git commit -m "Add backup files to gitignore"
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[master 1bb0ebf] Add backup files to gitignore
 1 file changed, 1 insertion(+), 1 deletion(-)

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git branch -a
git branch -d GitWork
git log --oneline --graph --decorate
GitWork
* master
  remotes/origin/master
Deleted branch GitWork (was b96402d).
* 1bb0ebf (HEAD -> master) Add backup files to gitignore
* 70d781e Resolve merge conflict between master and GitWork
| \
| * b96402d Add hello.xml in GitWork branch
* | f987618 Add hello.xml in master branch with different content
|/
* 863caa6 Add feature.txt in GitNewBranch
* 7392b8c (origin/master) Add gitignore file to ignore log files and folders
* 088c41c Welcome

AADI@Aadi MINGW64 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 1.68 KiB | 429.00 KiB/s, done.
Total 15 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Aadi4-gif/GitDemo.git
  7392b8c..1bb0ebf  master -> master

AADI@Aadi MINGW64 ~/GitDemo (master)
```


